

# Erdős Problem #111

EP-111

L1 - TRACTABLE

OPEN

Graph Theory

## ORIGINAL PROBLEM STATEMENT

If  $G$  is a graph let  $h_G(n)$  be defined such that any subgraph of  $G$  on  $n$  vertices can be made bipartite after deleting at most  $h_G(n)$  edges.

What is the behaviour of  $h_G(n)$ ? Is it true that  $h_G(n)/n \rightarrow \infty$  for every graph  $G$  with chromatic number  $\aleph_1$ ?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsignedMath>